# Criterion 1: Outcome-based Curriculum (120)

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# 1.1. Vision, Mission and Program Educational Objectives (PEOs) (40)

## 1.1.1. State the Vision and Mission of the Institute and the Department (05)

- **Purpose of Criterion**
  - **To ensure that the Institute and Department have well-defined Vision and Mission statements that align with national educational priorities, industry needs, and societal development.**
  - **These statements reflect the long-term aspirations (Vision) and present commitments (Mission) that guide all academic and administrative activities.**
  - **NBA assigns marks based on clarity, relevance, dissemination, and alignment of these statements with program outcomes.**
- **Vision Statement (Institute Level)**
  - **Definition: A concise declaration of the *future aspirations* of the institute – where it wants to be in 10–20 years.**
  - **Features:**
    - **Inspirational and forward-looking.**
    - **Addresses the role of the institute in education, research, and contribution to society.**
    - **Short, crisp, and easy to remember.**
  - **Example: *"To be a globally recognized centre of excellence in engineering education, research, and innovation for sustainable development of society."***

- **Mission Statement (Institute Level)**
  - **Definition: A statement of the current purpose and actions undertaken by the institute to achieve its vision.**
  - **Features:**
    - **Defines academic, research, and community service objectives.**
    - **Specifies the approach: teaching-learning process, industry collaboration, innovation, value-based education, etc.**
    - **More detailed than the vision.**
  - **Example:**
    - ***"To provide quality education through innovative pedagogy and advanced infrastructure."***
    - ***"To foster research and industry collaboration for real-world problem solving."***
    - ***"To nurture responsible professionals with ethics, leadership, and entrepreneurial skills."***

- ## Vision Statement (Department Level)
  - **Should align with the institute's vision but be discipline-specific.**
  - **Reflects the department's aspiration in terms of its core area (e.g., Computer Science, Mechanical Engineering).**
  - **Example: *"To be a leading department in producing globally competent computer science engineers capable of innovation and lifelong learning."***

- ## Mission Statement (Department Level)
  - **Departmental mission explains how the department will realize its vision, mapped to the discipline.**
  - **Includes aspects like curriculum design, research thrust areas, industry collaboration, student development.**
  - **Example:**
    - ***"To impart strong theoretical and practical knowledge in computing to meet industrial and societal needs."***
    - ***"To promote research, innovation, and entrepreneurship in emerging areas of computer science."***
    - ***"To instil professional ethics, teamwork, and continuous learning culture in graduates."***

- ## Supporting Evidence for NBA Documentation
  - **Officially approved Vision & Mission statements (signed by Governing Body/Academic Council).**
  - **Screenshots of website display of Vision/Mission.**
  - **Photographs of classroom/department boards where Vision & Mission are displayed.**
  - **Meeting minutes of Academic Council/IQAC/Departmental Board of Studies showing review/revision of Vision & Mission.**
  - **Mapping document showing how Vision/Mission aligns with PEOs, POs, and PSOs.**

- ## NBA Requirements for Marks

1. **Clarity and Appropriateness – Vision and Mission statements must be clear, relevant, and realistic.**
2. **Alignment – Department's vision/mission should align with the Institute's vision/mission.**
3. **Consistency – Should be in line with Program Educational Objectives (PEOs), Program Outcomes (POs), and Program Specific Outcomes (PSOs).**
4. **Dissemination – Statements should be displayed on the institute's website, handbooks, notice boards, and classrooms.**
5. **Stakeholder Involvement – Framing/reviewing vision and mission should involve faculty, students, alumni, industry experts, and management.**

# 1.1.2. State PEOs of the Program (05)

- **Purpose of PEOs in NBA Accreditation**
  - **PEOs describe the broad career and professional achievements that graduates are expected to accomplish within 3–5 years after graduation.**
  - **They act as a bridge between the Vision/Mission of the Department and the Program Outcomes (POs).**
  - **NBA evaluates whether PEOs are clearly stated, consistent with mission statements, and mapped to stakeholder needs (students, industry, alumni, society).**

- **Definition of PEOs**
  - **Long-term objectives describing what graduates will be able to achieve in their professional career, higher studies, entrepreneurship, or societal contribution.**
  - **PEOs are not course-level outcomes but program-level goals.**
  - **They are qualitative, broad in nature, and align with Bloom's higher levels (application, synthesis, evaluation).**

- **Key Features of Good PEOs**
  - **Must be few (3–5 statements) – not too many or too vague.**
  - **Should align with:**
    - **Department Vision and Mission.**
    - **Graduate attributes specified by NBA/ABET (e.g., engineering knowledge, problem analysis, design, ethics, teamwork).**
    - **National priorities (e.g., employability, innovation, sustainability).**
  - **Should be measurable indirectly via surveys (alumni, employer feedback, placement records).**

- **Supporting Evidence for NBA Documentation**
  - **PEO Statements (approved by Academic Council/BoS).**
  - **Mapping table showing linkage between Department Mission ↔ PEOs ↔ Program Outcomes (POs).**
  - **Process document of how PEOs were framed (committee meeting minutes with stakeholders).**
  - **Assessment reports of PEO attainment (alumni surveys, employer feedback).**
  - **Dissemination evidence – screenshots of website, program brochures, and handbooks.**

- **NBA Requirements for Marks**

1. **Clarity – PEOs must be clearly stated, specific, and realistic.**
2. **Alignment – PEOs should flow from the Vision & Mission of the Institute/Department.**
3. **Stakeholder Involvement – Framing PEOs should involve faculty, industry experts, alumni, and students.**
4. **Assessment & Evaluation Process – There should be a mechanism to assess PEO attainment (alumni surveys, employer surveys, placement data, higher studies data).**

- **Example PEOs (for Engineering Program)**

1. **Professional Competence – Graduates will excel in professional careers by applying fundamental engineering knowledge and modern tools.**
2. **Higher Studies & Lifelong Learning – Graduates will pursue higher education, research, and continuous professional development.**
3. **Innovation & Entrepreneurship – Graduates will demonstrate innovation, leadership, and entrepreneurial skills to develop solutions for societal needs.**
4. **Ethics & Social Responsibility – Graduates will practice professional ethics and contribute responsibly to sustainable development.**
5. **Teamwork & Communication – Graduates will work effectively in multidisciplinary teams and communicate efficiently in global environments.**

# 1.1.3. Process of Defining Vision, Mission and PEOs (15)

## 1. Purpose of this Criterion

- **NBA expects institutions not only to state Vision, Mission, and PEOs but also to document the systematic process through which they were defined.**
- **Marks are awarded based on:**
  - **Transparency of the process.**
  - **Stakeholder participation (faculty, students, alumni, industry, governing bodies).**
  - **Alignment with Institute Vision/Mission, national priorities, and program outcomes.**
  - **Review and continuous improvement mechanisms.**

## Process of Defining the Vision & Mission of the Department

- **Step 1 – Review Institute Vision & Mission**
  - **Departmental Vision/Mission must align with the larger goals of the institute.**
  - **Reference to statutory bodies like AICTE, UGC, NEP-2020 is considered.**
- **Step 2 – Internal Brainstorming**
  - **Faculty meetings to draft initial Vision/Mission reflecting:**
    - **Academic strengths of the department.**
    - **Future opportunities in the discipline.**
    - **Societal and industrial relevance.**
- **Step 3 – Stakeholder Consultation**
  - **Inputs are collected from:**
    - **Faculty members – academic feasibility.**
    - **Students – learning expectations.**
    - **Alumni – industry readiness & career development.**
    - **Industry experts – employability & skill needs.**
    - **Governing body/IQAC – institutional goals & quality assurance.**
- **Step 4 – Drafting Statements**
  - **Draft Vision (long-term aspirations) and Mission (current commitments).**
  - **Ensure statements are concise, clear, realistic, and discipline-specific.**
- **Step 5 – Approval & Dissemination**
  - **Final Vision/Mission approved by Board of Studies (BoS) and Academic Council.**
  - **Displayed on website, department notice boards, and program handbooks.**

# Process of Defining Program Educational Objectives (PEOs)

- **Step 1 – Identify Graduate Attributes/POs**
  - **Refer to NBA Graduate Attributes / Washington Accord competencies (knowledge, problem solving, ethics, lifelong learning).**
- **Step 2 – Map to Department Mission**
  - **Each PEO must support one or more departmental mission statements.**
- **Step 3 – Stakeholder Involvement**
  - **Faculty – curriculum strengths.**
  - **Alumni – career progression feedback.**
  - **Employers/Industry – workplace requirements.**
  - **Students – academic aspirations (higher studies, entrepreneurship, jobs).**
- **Step 4 – Drafting PEOs**
  - **Formulated as broad, measurable statements (3–5 in number).**
  - **Examples:**
    - **Graduates will excel in professional careers.**
    - **Graduates will pursue higher education and lifelong learning.**
    - **Graduates will demonstrate innovation, entrepreneurship, and leadership.**
    - **Graduates will uphold ethics and contribute to society.**

- **Step 5 – Approval & Documentation**
  - **Finalized by Board of Studies → approved by Academic Council.**
  - **Included in curriculum, website, program brochures.**
- **Step 6 – Review & Revision**
  - **Periodic review (every 3–5 years) based on feedback from:**
    - **Alumni surveys, employer surveys, accreditation audits, placement reports.**
  - **Updated to reflect emerging technologies, NEP-2020 guidelines, and industry needs.**

## Supporting Evidence for NBA Documentation

- **Minutes of Meetings – Department Advisory Committee, BoS, Academic Council.**
- **Stakeholder Feedback Forms – alumni, employers, students.**
- **Mapping Matrices – Vision ↔ Mission ↔ PEOs ↔ POs.**
- **Screenshots/Photos – Vision/Mission/PEOs displayed in classrooms, website.**
- **Review Reports – documentation of periodic updates.**

## NBA Requirements for Marks

1. **Stakeholder Participation – Process should involve internal and external stakeholders.**
2. **Systematic Approach – Clear steps from drafting → approval → dissemination.**
3. **Alignment – Department Vision/Mission ↔ Institute Vision/Mission ↔ PEOs ↔ POs.**
4. **Evidence of Review – Minutes of meetings, survey reports, BoS approvals, Academic Council resolutions.**
5. **Dissemination – Website, curriculum documents, classrooms, laboratories.**

# 1.1.4. Dissemination of Vision, Mission and PEOs (05)

## 1. Purpose of Dissemination

- **NBA emphasizes that Vision, Mission, and PEOs should not just exist on paper but be actively communicated to all stakeholders.**
- **Dissemination ensures that students, faculty, staff, alumni, employers, and society at large understand and align with the institute/departmental goals.**
- **Marks are awarded based on:**
  - **Visibility (multiple platforms of display).**
  - **Stakeholder Awareness (how information is shared and reinforced).**
  - **Effectiveness of communication (feedback mechanisms, awareness programs).**

## Dissemination Channels

- **Website (Institute + Departmental pages):**
  - **Vision, Mission, and PEOs prominently displayed on the home page or About Us/Academics section.**
  - **Digital brochures and handbooks downloadable with Vision/Mission/PEOs included.**
- **Curriculum Documents:**
  - **Program curriculum, syllabi, and academic regulations contain Vision, Mission, and PEOs at the beginning.**
  - **Student handbooks, course files, and faculty handbooks.**
- **Physical Display:**
  - **Posters, wall boards, flex boards in classrooms, laboratories, seminar halls, administrative offices, and library.**
  - **Printed in annual reports, newsletters, convocation booklets.**
- **Official Documents & Reports:**
  - **IQAC reports, accreditation reports, program files, and student induction booklets.**
  - **Mentioned in MoUs, collaborations, and funding proposals to show alignment with societal needs.**
- **Orientation & Induction Programs:**
  - **Introduced to new students, faculty, and staff during orientation sessions.**
  - **Awareness workshops conducted regularly to reinforce these statements.**

# Internal Stakeholder Awareness

- **Management & Governing Board:**
  - **Vision, Mission, and PEOs shared in governing body meetings, IQAC, and Board of Studies meetings.**
- **Faculty & Staff:**
  - **Discussed in faculty meetings, departmental meetings, training workshops.**
  - **Faculty encouraged to integrate Vision/Mission into lesson plans, teaching, and mentoring.**
- **Students:**
  - **Explained in student induction, class orientations, mentoring sessions.**
  - **Displayed in classrooms and labs for continuous visibility.**
  - **Reinforced via student projects, hackathons, and co-curricular activities.**

# External Stakeholder Awareness

- **Employers & Industry Experts:**
  - **Shared during Industry-Institute Interaction Cell (IIIC) meetings, placement drives, MoUs, industrial visits.**
- **Alumni:**
  - **Communicated via alumni meets, newsletters, LinkedIn groups, and feedback surveys.**
- **Funding Agencies & Professional Bodies:**
  - **Included in project proposals, consultancy agreements, and research grant applications.**
- **Parents & Society:**
  - **Highlighted during parent-teacher meetings, cultural events, annual day reports.**

## Process to Ensure Effective Implementation

- **Step 1: Draft Vision, Mission, and PEOs → Approval by Board of Studies/Academic Council.**
- **Step 2: Publish in official documents → Website, curriculum, student handbook.**
- **Step 3: Display physically → Posters/boards in institute and department.**
- **Step 4: Awareness drives → Orientation, workshops, faculty-student meetings.**
- **Step 5: Reinforcement → Continuous linkage to Program Outcomes (POs) in teaching-learning.**
- **Step 6: Feedback & Review → Periodic surveys from students, alumni, employers to check awareness and relevance.**

# Supporting Evidence for NBA Documentation

- **Screenshots of website pages showing Vision, Mission, PEOs.**
- **Photographs of classrooms, labs, and offices with posters/flex boards.**
- **Extracts from curriculum documents, student/employee handbooks.**
- **Induction program schedules showing sessions on Vision & Mission.**
- **Minutes of meetings (BoS, IQAC, Alumni Meet) where Vision/Mission/PEOs were presented.**
- **Survey reports (student/alumni/employer feedback) validating stakeholder awareness.**

# 1.1.5. Mapping of PEOs with Mission (10)

## Purpose of Mapping

- **NBA requires every program to demonstrate alignment between the Mission of the Department and the Program Educational Objectives (PEOs).**
- **This mapping shows how each Mission element contributes to achieving long-term PEOs.**
- **Marks are awarded for:**
  - **Clarity of Mission breakdown (M1, M2, … Mn).**
  - **Logical mapping of each PEO with relevant Mission elements.**
  - **Justification of correlation levels (Low–1, Medium–2, High–3).**

## Step-by-Step Process

- **Step 1 – Define Mission Elements**
  - **Break down the Department Mission into distinct measurable elements (M1, M2, M3…).**
  - **Example (Computer Science Department):**
    - **M1: Provide strong theoretical and practical foundations in computing.**
    - **M2: Foster research, innovation, and entrepreneurship.**
    - **M3: Instill professional ethics, teamwork, and communication skills.**
    - **M4: Collaborate with industry for real-world problem-solving.**
- **Step 2 – Define PEOs**
  - **PEO1: Graduates will excel in professional careers through strong computing knowledge.**
  - **PEO2: Graduates will pursue higher education, research, and lifelong learning.**
  - **PEO3: Graduates will demonstrate innovation and entrepreneurship.**
  - **PEO4: Graduates will practice ethical values and contribute to society.**
- **Step 3 – Create the Mapping Matrix**
  - **Table format given by NBA:**
  - **Use correlation levels Low (1), Medium (2), High (3).**
  - **"–" if no correlation.**

## Sample Matrix

## Mapping of PEOs with Department Mission

| PEO Statements | M1 | M2 | M3 | M4 | Justification |
|---|---|---|---|---|---|
| PEO1: Graduates will excel in professional careers through strong computing knowledge | 3 | 2 | 1 | 2 | M1 directly builds strong computing foundation → high correlation. M2 research contributes moderately. M3 teamwork adds basic support. M4 industry collaboration ensures applicability. |
| PEO2: Graduates will pursue higher education, research, and lifelong learning | 2 | 3 | 1 | 2 | M2 research focus highly supports higher education. M1 knowledge provides moderate foundation. M4 industry connects to advanced exposure. |
| PEO3: Graduates will demonstrate innovation and entrepreneurship | 1 | 3 | 1 | 2 | M2 directly emphasizes innovation/entrepreneurship. M4 ensures exposure to industry challenges. |
| PEO4: Graduates will practice ethical values and contribute to society | 1 | 1 | 3 | 1 | M3 ethics/teamwork directly supports this objective (high). M1/M2/M4 contribute indirectly at low level. |

# NBA Requirements for Full Marks

1. Clearly defined Mission elements (M1…Mn).
2. Well-written PEOs linked logically to Mission.
3. Matrix table filled with Low/Medium/High correlation (not arbitrary, but justified).
4. Narrative justification for each mapping row (not just numbers).
5. Evidence of stakeholder involvement in preparing mapping (BoS/IQAC minutes).

# 1.2. Curriculum Structure and Features (30)

## 1.2.1. Program Curriculum Structure (05)

**Purpose of Curriculum Structure in NBA**

- **NBA requires programs to present the overall structure of the curriculum in terms of:**
  - **Teaching and Learning Scheme (L–T–P distribution).**
  - **Credits allocated to each course and category.**
  - **Balance between basic sciences, engineering sciences, core courses, electives, projects, and humanities.**
- **Ensures alignment with AICTE/NBA model curriculum and Washington Accord graduate attributes/POs.**

## Components of Curriculum Structure

### A. Course Categories

1. Basic Science Courses (BSC): Mathematics, Physics, Chemistry, etc.
2. Engineering Science Courses (ESC): Workshop, Engineering Graphics, Programming, Basic Electrical/Electronics.
3. Professional Core (PCC): Core courses specific to the discipline.
4. Professional Electives (PEC): Department-specific electives chosen by students.
5. Open Electives (OEC): Courses offered by other departments to broaden knowledge.
6. Humanities and Social Sciences (HSMC): Communication skills, Management, Ethics.
7. Projects / Mini Projects / Capstone: Industry projects, research-based assignments.
8. Internship / Industry Training: As per AICTE/NEP-2020 guidelines.

### B. Teaching–Learning Scheme

- Each course is defined in L–T–P format:
  - L: Lecture hours/week
  - T: Tutorial hours/week
  - P: Practical/Lab hours/week
- Credit Calculation Rule (AICTE/NBA):
  - 1 Lecture hour/week = 1 Credit
  - 1 Tutorial hour/week = 1 Credit
  - 2 Practical/Lab hours/week = 1 Credit

# Example Curriculum Structure Table

| Course Category | L | T | P | Credits | Remarks |
|---|---|---|---|---|---|
| Basic Science (BSC) | 12 | 2 | 6 | 20 | Physics, Chemistry, Mathematics |
| Engineering Science (ESC) | 8 | 0 | 6 | 14 | Engineering Graphics, Programming, Workshop |
| Professional Core (PCC) | 28 | 4 | 12 | 44 | Core departmental subjects |
| Professional Electives (PEC) | 6 | 0 | 4 | 10 | Specialized electives in chosen area |
| Open Electives (OEC) | 6 | 0 | 2 | 8 | Electives from other departments |
| Humanities & Management (HSMC) | 6 | 2 | 2 | 10 | Communication, Ethics, Management |
| Projects / Seminar | 0 | 0 | 12 | 6 | Mini/Capstone project |
| Internship / Industrial Training | – | – | – | 4 | Mandatory internship |
| Total | – | – | – | 120–160 | (As per AICTE/NBA norms) |

## Format of Teaching & Learning Scheme Table

| Course Code | Course Title | Classroom Instruction (CI) (hrs/sem) | Lab Instruction (LI) (hrs/sem) | Term Work (TW) + Self Learning (SL) (hrs/sem) | Total Hours (CI+LI+TW+SL) | Credits (Total Hours ÷ 30) |
|---|---|---|---|---|---|---|
| 101 | C++ Programming | 42 | 14 | 28 + 36 | 120 | 120 ÷ 30 = 4 |
| 102 | Chemistry Laboratory | 0 | 42 | 18 | 60 | 60 ÷ 30 = 2 |
| 103 | Engineering Mathematics | 56 | 0 | 28 | 84 | 84 ÷ 30 = 3 |
| 104 | Electrical Workshop | 28 | 28 | 14 | 70 | 70 ÷ 30 ≈ 2 |
| 105 | Communication Skills | 28 | 0 | 30 | 58 | 58 ÷ 30 ≈ 2 |
| … | … | … | … | … | … | … |

## Explanation of Columns

- **CI (Classroom Instruction):**
  Includes **lectures, tutorials, case discussions, problem-based learning, video demos, etc.**
- **LI (Laboratory Instruction):**
  Includes **practicals, experiments, workshops, field visits, live demonstrations**.
- **TW (Term Work):**
  Includes **assignments, seminars, micro-projects, industrial visits, group work**.
- **SL (Self Learning):**
  Includes **MOOCs, NPTEL, Spoken Tutorials, online platforms, flipped classroom prep, extra readings**.
- **Total Hours:**
  Sum of CI + LI + TW + SL.
- **Credits:**
  Computed as **Total Hours ÷ 30**, in line with the **National Credit Framework (NCF)**.
  (For universities still using L–T–P system, hours must first be converted into equivalent totals).

## Documentation Required in SAR

- **Curriculum Structure Table (as above, program-wise).**
- **Semester-wise Course Distribution Sheet (showing L–T–P–C for each course).**
- **Credit Summary Sheet (category-wise).**

**Mapping to NBA Graduate Attributes (POs/WKs).**

## NBA Documentation Tip

- **Include semester-wise tables for *all courses in the program*.**
- **Show course-wise credit calculation as per NCF.**
- **Attach Annexure with complete program teaching–learning scheme (all 8 semesters for UG).**

# 1.2.2. Components of Program Curriculum (05)

## Purpose of this Criterion

- NBA evaluates whether the program curriculum is well balanced across various categories:
  - Sciences, Engineering fundamentals, Core subjects, Electives, Humanities, Projects, and Internships.
- Ensures the curriculum reflects AICTE Model Curriculum / Washington Accord guidelines.
- Proves that students are not only trained in technical knowledge but also in professional, ethical, and lifelong learning skills.

## Standard Components of Curriculum

1. Basic Sciences (BSC): Mathematics, Physics, Chemistry, Environmental Science.
2. Basic Engineering Sciences (ESC): Engineering Graphics, Workshop, Programming, Electrical/Mechanical Fundamentals.
3. Humanities & Social Sciences (HSMC): Communication Skills, Management, Economics, Ethics.
4. Program Core (PCC): Discipline-specific core courses that form the foundation of the program.
5. Program Electives (PEC): Specialized courses within the discipline for advanced learning.
6. Open Electives (OEC): Interdisciplinary courses chosen from other departments.
7. Projects: Mini projects, final-year major project, capstone.
8. Internships/Seminars: Industry training, internships, technical seminars, research paper presentations.
9. Any Other (as per Institute): MOOCs, value-added courses, non-credit audit courses, entrepreneurship modules.

## Example SAR Table – Program Curriculum Components

**Program Curriculum Grouping Based on Components**

| Curriculum Component | % of Total Credits | Total Contact Hours | Total Credits |
|---|---|---|---|
| Basic Sciences (BSC) | 15% | 480 | 24 |
| Basic Engineering Sciences (ESC) | 10% | 320 | 16 |
| Humanities & Social Sciences | 8% | 240 | 12 |
| Program Core (PCC) | 40% | 1280 | 48 |
| Program Electives (PEC) | 10% | 320 | 12 |
| Open Electives (OEC) | 5% | 160 | 8 |
| Projects (Mini + Major) | 7% | 240 | 8 |
| Internships/Seminars | 5% | 160 | 6 |
| Any Other (MOOCs/Value Added) | – | 80 | 2 |
| Total | 100% | 3280 | 126 |

## Interpretation

- **Balance of Credits:**
  - **15–20% Basic Sciences & Engineering → foundation.**
  - **~40% Program Core → discipline expertise.**
  - **15–20% Electives (Program + Open) → flexibility & interdisciplinary exposure.**
  - **~10–12% Projects/Internships → hands-on learning & industry readiness.**

**Remaining % → Humanities, MOOC, value-added courses.**

## NBA Documentation Tip

- **Present year/semester-wise curriculum sheets with L–T–P–C structure.**
- **Attach consolidated grouping table as above**
- **Cross-reference with AICTE/NBA model guidelines to show compliance.**
- **Use graphs/pie charts in annexure to show percentage distribution of credits.**

## 1.2.3. State the Process Used to Identify Extent of Compliance of the University Curriculum for Attaining the Program Outcomes and Program Specific Outcomes as mentioned in Annexure II. Also Mention the Identified Curricular Gaps, if any (10)

### Objective

- **To ensure that the existing University-prescribed curriculum adequately supports the attainment of all Program Outcomes (POs) and Program Specific Outcomes (PSOs).**
- **To identify curricular gaps that may hinder the holistic development of students' technical, professional, and ethical competencies.**

**To strengthen the Outcome-Based Education (OBE) framework by integrating academic, industrial, and research-oriented perspectives**

## Methodology Adopted for Compliance Evaluation

- **1. Curriculum–Outcome Mapping Framework**
  - **Each Course Outcome (CO) of every subject is mapped to relevant POs and PSOs as per NBA guidelines.**
  - **Mapping matrix prepared using a 3-point scale (1–Low, 2–Medium, 3–High) indicating the strength of correlation.**
  - **Faculty coordinators and OBE committee perform mapping through departmental academic review meetings.**
- **2. Review through Board of Studies (BoS) and Department Academic Committee (DAC)**
  - **Regular evaluation of University syllabus by departmental experts, industry representatives, and alumni members.**
  - **Discussion on adequacy of topics for achieving cognitive (knowledge), affective (skills), and psychomotor (application) domains.**
  - **Recommendations forwarded to the University through the Academic Council for inclusion of advanced or interdisciplinary content.**
- **3. Stakeholder Feedback Analysis**
  - **Students: Feedback collected on curriculum relevance and applicability to industry and higher studies.**
  - **Alumni: Input on alignment between academic learning and professional requirements.**
  - **Employers: Suggestions on new technologies, tools, or skills to enhance employability.**
  - **Faculty: Academic audit of syllabus adequacy for achieving specific outcomes.**
  - **Consolidated feedback analyzed by IQAC/OBE cell to identify missing components.**
- **4. Gap Identification through Bloom's Taxonomy and Attainment Analysis**
  - **Learning levels (Remember, Understand, Apply, Analyze, Evaluate, Create) mapped for all COs.**
  - **If outcomes predominantly fall in lower-order skills (R/U/A), curriculum gaps are marked for higher-order reinforcement (An/Ev/Cr).**
  - **Annual comparison of attainment levels vs target levels to locate deficiencies.**
- **5. Benchmarking and External Reference Studies**
  - **Curriculum compared with top-tier universities (IITs, NITs) and global standards (ABET, Washington Accord).**
  - **Industry 4.0 alignment check—emerging domains like AI, Cybersecurity, IoT, Cloud, Sustainability, and Data Science.**
  - **Identification of topics absent in current syllabus but vital for 21st-century engineering competencies.**

# Identified Curricular Gaps

| Gap Type | Description / Observation | Affected POs/PSOs | Action Taken / Proposed |
|---|---|---|---|
| Technological Gap | Limited exposure to AI/ML, IoT, and Data Analytics tools. | PO5, PSO3 | Introduced value-added courses on Python, ML, and IoT; Guest lectures by industry experts. |
| Practical Exposure Gap | Inadequate industry-oriented lab experiments and case studies. | PO3, PO11 | Added mini projects beyond syllabus; organized internships and industrial visits. |
| Soft Skills & Communication Gap | Less emphasis on technical writing and oral communication. | PO10 | Conducted communication skill workshops and project presentation sessions. |
| Sustainability and Ethics Gap | Few topics related to SDGs, environmental ethics, and green technologies. | PO7, PO8 | Introduced seminars on sustainability, energy audits, and ethical engineering practices. |
| Interdisciplinary Learning Gap | Lack of cross-domain exposure (e.g., AI in Healthcare, Data Analytics in Civil Engineering). | PO12, PSO2 | Initiated interdisciplinary workshops and elective subjects from other departments. |
| Entrepreneurship & Innovation Gap | Limited coverage on innovation management and startup ecosystem. | PO9, PSO3 | Collaborated with IIC for innovation challenges, entrepreneurship development cells. |

## Corrective Measures and Continuous Improvement Mechanisms

- **Curriculum Enrichment Beyond Syllabus**
  - **Add-on courses, internships, MOOC certifications, hackathons, and research projects integrated each semester.**
- **Department Academic Audit**
  - **Internal audit once per semester to verify CO–PO–PSO coverage, course files, and assessment rubrics.**
- **Integration with University and BoS**
  - **Department submits curriculum revision proposals to BoS with identified gap reports annually.**
- **IQAC/OBE Review**
  - **IQAC conducts post-semester reviews comparing attainment levels and recommending enrichment programs.**
- **Outcome Re-Alignment**
  - **Revised mapping ensures even distribution of attainment across cognitive levels and outcome domains.**

## Evidence and Documentation Maintained

- **CO–PO–PSO Mapping Matrix for all courses.**
- **Stakeholder feedback reports and analysis charts.**
- **Departmental Academic Audit Reports.**
- **Action Taken Reports (ATR) on identified gaps.**
- **University correspondence for syllabus modification proposals.**
- **Comparative benchmarking matrix with national and international curricula.**

# Curricular Gap Identification and Action Plan

| Gap ID | Gap Type | Description of Gap | Affected POs/PSOs | Source of Identification | Action Proposed | Action Implemented (Year) | Supporting Evidence (File No.) |
|---|---|---|---|---|---|---|---|
| G1 | Technological | Lack of AI/ML exposure in core subjects | PO5, PSO3 | Employer feedback | Introduce add-on course on ML | Implemented 2024–25 | GAP_AI_COURSE_2025.pdf |
| G2 | Practical | Less lab exposure for Data Analytics | PO3, PSO1 | Faculty review | Conduct workshop on R/Python | Implemented 2023–24 | GAP_LAB_TRAINING_2024.pdf |
| G3 | Communication | Weak technical report writing | PO10 | Alumni survey | Add seminar on technical writing | Ongoing | GAP_WRITING_2025.pdf |
| G4 | Sustainability | Few topics on SDG-based projects | PO7 | IQAC review | Organize SDG innovation challenge | Planned 2025–26 | GAP_SDG_EVENT_2025.pdf |

# 1.2.4. State the Delivery Details of the Content beyond the Syllabus for the Attainment of Program Outcomes and Program Specific Outcomes (10)

## Objective and Rationale

- **Purpose:**
  - **To strengthen and expand students' learning experience by integrating contemporary, interdisciplinary, and industry-relevant topics beyond the university curriculum.**
  - **To ensure that the graduates attain not only the cognitive competencies mentioned in the syllabus but also practical, analytical, and societal capabilities aligned with NBA-defined Program Outcomes (POs) and Program Specific Outcomes (PSOs).**
- **Rationale:**
  - **Rapid technological advancements often outpace curricular revisions; therefore, inclusion of beyond-syllabus content bridges the curriculum–industry gap.**
  - **It promotes lifelong learning, innovation, social responsibility, and entrepreneurial thinking, which are vital for outcome-based education (OBE).**

# Modes of Delivery and Implementation Strategies

- **1. Guest Lectures / Expert Interactions**
  - **Organized every semester through IQAC and Department-level Academic Calendar.**
  - **Resource persons from academia, industry, and R&D labs deliver sessions on emerging topics.**
  - ***Examples:***
    - **"Applications of Deep Learning in Healthcare" – maps to PO5 (Modern tool usage), PO12 (Lifelong learning).**
    - **"Energy Audit in Smart Grids" – maps to PO7 (Environment & Sustainability), PSO2 (Renewable energy systems).**
  - ***Documentation:*** **Event report, expert profile, participant list, photos, feedback analysis, and PO mapping chart.**
- **2. Workshops / Hands-on Training Sessions**
  - **Conducted in collaboration with technical societies (IEEE, CSI, SAE, IIC, etc.).**
  - **Duration ranges from 1 day to 1 week, with practical lab exercises.**
  - ***Examples:***
    - **"3D Modelling and Simulation using SolidWorks" – maps to PO3, PSO1.**
    - **"Python for Data Analytics" – maps to PO4, PO5, PSO3.**
  - ***Outcome Measurement:*** **Pre/post assessment test, skill certification, and project submission.**

- **3. Add-on / Certificate Courses**
  - **Value-added courses conducted during semester breaks or weekends, recognized by the institute or external bodies (Coursera, IIT NPTEL, AICTE-ATAL, etc.).**
  - ***Examples:***
    - **"Cloud Computing with AWS" → PO5, PO11.**
    - **"Entrepreneurship and Innovation Management" → PO9, PO10.**
  - ***Impact:*** **Improves employability skills, enhances placement ratio, and supports higher PO attainment.**
- **4. Seminars, Webinars, and Panel Discussions**
  - **Thematic sessions on socio-technical issues, ethical engineering, sustainability, and global trends.**
  - ***Examples:***
    - **"Role of Engineers in Achieving SDGs" → PO6, PO7.**
    - **"AI Ethics and Responsible Innovation" → PO8, PO12.**
  - ***Assessment:*** **Student feedback, participation certificates, reflective summary reports.**
- **5. Industrial Visits / Field Learning**
  - **Organized for each year to expose students to real-world industrial processes and applications.**
  - ***Examples:***
    - **Visit to manufacturing units, renewable energy plants, data centers.**
    - ***PO Mapping:*** **PO6 (Societal impact), PO10 (Communication skills).**

***Evidence:*** **Visit report, student reflection sheet, photographic documentation, employer interaction feedback.**

- **6. Mini Projects / Capstone Beyond Curriculum**
  - **Students undertake short research or innovation-based projects using new technologies not covered in the syllabus.**
  - ***Examples:***
    - **"IoT-based Smart Irrigation System" → PO3, PSO1.**
    - **"AI Chatbot for Campus Support" → PO5, PSO3.**
  - ***Evaluation:*** **Rubric-based project evaluation, mentor assessment, and presentation before a review committee.**
- **7. Innovation, Ideathon, and Hackathon Participation**
  - **Institutional Innovation Cell (IIC) conducts intra/inter-college events encouraging innovative problem-solving.**
  - ***Examples:***
    - **"Smart India Hackathon", "AICTE IDEA Lab Competitions."**
  - ***Mapped POs:*** **PO8 (Ethics), PO9 (Teamwork), PO11 (Project Management).**
  - ***Outcome:*** **Enhanced creativity, teamwork, and application of knowledge beyond theory.**
- **8. Research Paper Writing and Publication Workshops**
  - **Encourages students to publish papers in Scopus/UGC indexed journals or conferences.**
  - **Sessions cover literature survey, plagiarism checks, reference management, and academic writing ethics.**
  - ***PO Mapping:*** **PO4 (Investigation of complex problems), PO12 (Lifelong learning).**
  - ***Output:*** **Research publications, awards, and participation in conferences.**
- **9. Social Outreach and Extension Activities**
  - **Programs under NSS, Unnat Bharat Abhiyan, and SDG clubs include technology-based social problem-solving.**
  - ***Examples:***
    - **"Low-cost water filtration design for rural communities."**
    - ***Mapped POs:*** **PO6 (Societal impact), PO7 (Environment and sustainability).**

***Impact:*** **Instills civic responsibility and ethical awareness among students**

# Documentation and Evidence Framework

| Activity | Topic/Title | Mode of Delivery | Resource Person/Organization | Duration | Targeted POs/PSOs | Assessment Method | Evidence (Doc/File No.) |
|---|---|---|---|---|---|---|---|
| Guest Lecture | "AI in Medical Imaging" | Lecture + Demo | Dr. XYZ, Philips R&D | 1 Day | PO5, PSO2 | Quiz + Feedback | 1.2.4_GL_AIMI_2025.pdf |
| Workshop | "IoT and Embedded Systems" | Hands-on | Edgate Technologies | 5 Days | PO3, PSO1 | Project Demo | 1.2.4_WS_IOT_2025.xlsx |
| Add-on Course | "Digital Marketing Analytics" | Online (Coursera) | Google Academy | 4 Weeks | PO9, PSO3 | Certificate + Test | 1.2.4_CO_DMA_2025.pdf |
| Field Visit | "Solar Power Plant" | Industrial Visit | Tata Power Solar | 1 Day | PO6, PO7 | Report | 1.2.4_IV_SP_2025.pdf |

# Details of Events Organized to Cover Content Beyond the Syllabus

| S No. | PO/PSO Addressed / Gap Identified | Name of the Event | Date of Event | Resource Person / Organization | Relevance to POs/PSOs | CAY | Evidence File No. / Link |
|---|---|---|---|---|---|---|---|
| 1 | PO5 – Modern Tool Usage Gap | Workshop on "Python for AI Applications" | 12 Aug 2024 | Mr. Ramesh Kumar, Infosys Ltd. | Enhances programming and AI skills → PO5, PSO3 | CAY | EVT_AI_PY_2024.pdf |
| 2 | PO7 – Environment & Sustainability | Seminar on "Green Technologies for Smart Cities" | 05 Jan 2023 | Dr. Priya Menon, TERI Institute | Promotes SDG alignment → PO7, PO12 | CAYm1 | EVT_GT_2023.pdf |
| 3 | PSO2 – IoT Applications | Hackathon on "Smart Agriculture Solutions" | 18 Sep 2022 | IIC & MSME Innovation Cell | Strengthens problem solving & innovation → PO3, PO9, PSO2 | CAYm2 | EVT_SA_2022.pdf |
| 4 | PO10 – Communication Skills | Technical Paper Presentation Contest | 22 Mar 2021 | Department Faculty Panel | Builds communication & reporting skills → PO10 | CAYm3 | EVT_TPP_2021.pdf |

# 1.3. PO, PSO and their Mapping with Courses (15)

## 1.3.1. POs and PSOs (05)

### Program Outcomes (POs)

- **Definition:**
  - **POs are graduate attributes expected of every engineering graduate irrespective of discipline.**
  - **NBA defines 12 standard POs, aligned with Washington Accord (WKs).**
- **Purpose:**
  - **Ensure graduates have the required knowledge, skills, and attitudes.**
  - **Serve as a benchmark for Outcome-Based Education (OBE).**

**List of POs (As per NBA, Annexure II)**

1. **PO1 – Engineering Knowledge: Apply knowledge of mathematics, science, and engineering fundamentals to solve complex problems.**
2. **PO2 – Problem Analysis: Identify, formulate, and analyze problems using principles of mathematics, science, and engineering.**
3. **PO3 – Design/Development of Solutions: Design solutions for complex engineering problems considering safety, environment, and societal factors.**
4. **PO4 – Investigations of Complex Problems: Conduct experiments, analyze and interpret data, and provide valid conclusions.**
5. **PO5 – Modern Tool Usage: Create, select, and apply modern tools, software, and techniques with an understanding of their limitations.**
6. **PO6 – The Engineer & Society: Apply reasoning to assess societal, health, legal, and cultural issues relevant to professional practice.**
7. **PO7 – Environment & Sustainability: Understand impact of engineering solutions in societal/environmental contexts and need for sustainable development.**
8. **PO8 – Ethics: Apply ethical principles and commit to professional ethics, norms, and responsibilities.**
9. **PO9 – Individual & Team Work: Function effectively as an individual and in diverse/multidisciplinary teams.**
10. **PO10 – Communication: Communicate effectively with society, peers, and industry; prepare technical reports and presentations.**
11. **PO11 – Project Management & Finance: Demonstrate knowledge of management and finance principles in project work.**
12. **PO12 – Life-long Learning: Recognize the need for independent learning in technological change contexts.**

## Program Specific Outcomes (PSOs)

- **Definition:**
  - **PSOs are discipline-specific competencies a graduate of a particular program should attain.**
  - **Limited to 3 well-defined PSOs per program.**
- **Purpose:**
  - **Highlight unique skills graduates of this program will have (beyond general POs).**
  - **Derived from program vision, mission, industry needs, and stakeholder inputs.**

**Examples (for a Computer Science & Engineering Program)**

- **PSO1 – Core Competency: Graduates will be able to apply principles of computer science, programming, and mathematics to develop efficient software solutions.**
- **PSO2 – Emerging Technologies & Research: Graduates will demonstrate skills in AI, Data Science, and cloud technologies with the ability to engage in research and innovation.**
- **PSO3 – Professional Application: Graduates will apply computing knowledge to multidisciplinary domains such as healthcare, finance, and e-governance, adhering to ethical and sustainable practices.**

***(For other programs, PSOs should be tailored, e.g., Electrical Engg → Renewable Energy, Electronics → VLSI/Embedded Systems, Mechanical → CAD/CAM/Robotics, etc.)***

# 1.3.2. Mapping between the Courses and POs/PSOs (10)

## Purpose

- To show how each course contributes to NBA's 12 Program Outcomes (POs).
- Traceability: Course Outcomes (COs) → Program Outcomes (POs).
- Used for attainment calculation later (from student performance).

## Methodology

1. Define Course Outcomes (COs) for each course (4–6 per course).
2. Map each CO to relevant POs with correlation levels:
   - 3 = High, 2 = Medium, 1 = Low, – = No correlation.
3. Prepare course-wise tables (one per course).
4. Consolidate into a summary sheet (all courses → all POs).

## Example – Course: Data Structures (CS301)

| Course Outcomes (COs) | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1: Explain and analyze data structures (arrays, linked lists, trees, graphs). | 3 | 3 | 2 | – | – | – | – | – | – | – | – | – |
| CO2: Apply data structures in problem solving using algorithms. | 2 | 3 | 3 | 2 | – | – | – | – | – | – | – | – |
| CO3: Implement programs for real-world applications. | – | 2 | 3 | 3 | 2 | – | – | – | – | – | – | – |
| CO4: Evaluate efficiency of algorithms and optimize solutions. | – | 2 | 3 | 3 | 3 | – | – | – | – | – | – | – |

Legend:

- 3 = High Correlation
- 2 = Medium Correlation
- 1 = Low Correlation
- – = No Correlation

# 1.4. Course Outcomes and Course Articulation Matrix (25)

## 1.4.1. Course Outcome (Semester Wise) (10)

Purpose (Why NBA asks for COs semester-wise)

- To demonstrate how each course contributes to measurable learning outcomes (COs).
- Shows alignment of COs → POs → PSOs → PEOs → Vision/Mission.
- Evaluators want semester-wise evidence that curriculum is outcome-based.

Guidelines for Defining COs

- Maximum 4–6 COs per course.
- Use action-oriented verbs (Bloom's Taxonomy): Apply, Analyze, Design, Evaluate, Implement.
- Must be measurable (can be assessed via exams, projects, assignments).
- Should clearly map to POs and PSOs.

**Course Title:** Engineering Mathematics I
**Course Code:** MA101

| CO No. | Course Outcome (CO) | Mapped POs | Mapped PSOs |
|---|---|---|---|
| CO1 | Apply differential calculus to solve engineering problems. | PO1 | PSO1 |
| CO2 | Solve linear algebra problems using matrices and determinants. | PO1 | PSO1 |
| CO3 | Interpret basic calculus principles in engineering applications. | PO2 | – |
| CO4 | Use mathematical modeling to analyze real-life scenarios. | PO2, PO4 | PSO1 |

## NBA Tips

- **Provide 2 core courses per semester (I–VIII) → ~16 courses in total.**
- **Keep 4–6 COs per course, written in measurable terms.**
- **Ensure coverage across all 12 POs + PSOs.**
- **Cross-reference with mapping matrices.**
- **Use Bloom's verbs (apply, analyze, design, implement, evaluate).**

# 1.4.2. Course Articulation Matrix (15)

- **Definition:**
  - **A matrix table that shows the extent to which each CO contributes to POs and PSOs.**
  - **Ensures COs are not isolated but contribute to graduate attributes.**
- **Structure (Table 1.4.2):**
  - **Rows = Course Outcomes (CO1–CO6).**
  - **Columns = POs (1–12) + PSOs (up to 3).**
  - **Cell Values = Correlation levels:**
    - **3 = High (Substantial)**
    - **2 = Medium (Moderate)**
    - **1 = Low (Slight)**
    - **– = No correlation**

# Sample (Course: Data Structures, CS301):

| CO → / PO & PSO ↓ | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1: Explain & analyze data structures | 3 | 3 | – | – | – | – | – | – | – | – | – | – | 3 | – | – |
| CO2: Apply DS to algorithms | 2 | 3 | 3 | 2 | – | – | – | – | – | – | – | – | 3 | – | – |
| CO3: Implement programs | – | 2 | 3 | 3 | 3 | – | – | – | – | – | – | – | 3 | 2 | – |
| CO4: Evaluate efficiency | – | 3 | 3 | 3 | 2 | – | – | – | – | – | – | – | – | 3 | – |

**NBA Tips for Full Marks**

- Cover **two representative core courses per semester (16 courses total)**.
- Define **4–6 COs per course**.
- Keep articulation **balanced and realistic** (not all 3s).
- Ensure all **12 POs and PSOs** are covered across semesters.
- Provide **justification notes** (why a CO has strong mapping to a PO/PSO).

Attach **syllabus extracts, lab manuals, and project briefs** as evidence

# 1.5. Program Articulation Matrix (10)

## Purpose

- The Program Articulation Matrix (PAM) consolidates how all courses (through their COs) contribute to the 12 NBA-defined POs and the program's PSOs.
- Unlike the Course Articulation Matrix (CO ↔ PO/PSO) which is at the course level, this is a program-level summary.
- It demonstrates comprehensive coverage of all POs/PSOs across the curriculum.

## Methodology

1. Each course defines its CO–PO mapping (already done in Course Articulation Matrix).
2. Aggregate all course-level contributions to form a program-wide mapping.
3. Indicate correlation levels:
   - 3 = High (Substantial) → Course contributes strongly to that PO.
   - 2 = Medium (Moderate) → Course has partial contribution.
   - 1 = Low (Slight) → Course has minor contribution.
   - – = No contribution.
4. Ensure all 12 POs and PSOs are addressed by at least one course.

# Model Table – Program Articulation Matrix

| Course Code | Course Name | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| C101 | Engineering Mathematics I | 3 | 2 | – | – | – | – | – | – | – | – | – | – | 3 | – | – |
| C202 | Basic Electrical Engg. | 3 | 3 | 2 | 2 | 2 | – | – | – | – | – | – | – | 3 | – | – |
| C303 | Data Structures | 3 | 3 | 3 | 2 | 3 | – | – | – | – | – | – | – | 3 | 2 | – |
| C404 | Operating Systems | 3 | 3 | 3 | 3 | 2 | – | – | – | – | – | – | – | 3 | 3 | 2 |
| C505 | Software Engineering | 2 | 3 | 3 | 2 | 3 | – | – | – | 3 | 3 | 3 | – | 3 | 3 | 3 |
| C606 | Computer Networks | 3 | 3 | 2 | 3 | 3 | – | 2 | 2 | – | 2 | – | 2 | 3 | 3 | 2 |
| C707 | Artificial Intelligence | 3 | 3 | 3 | 3 | 3 | – | – | – | – | – | – | 3 | 3 | 3 | 3 |
| C808 | Major Project | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |

## Interpretation

- Breadth: Each PO and PSO is mapped to multiple courses (not just one).
- Progression: Lower semester courses → more PO1, PO2 (knowledge, analysis).
- Higher semester courses → more PO3, PO4, PO5 (design, investigation, modern tools), plus PO6–PO12 (ethics, teamwork, management, lifelong learning).
- Capstone (Major Project) usually covers all POs and PSOs at High (3) level.

## NBA Tips

- Ensure all 12 POs + PSOs are covered (no blank columns).
- Avoid unrealistic mappings (not all “3”).
- Show logical alignment:
  - Core science/engineering → PO1, PO2.
  - Labs/mini projects → PO3, PO4, PO5, PO9.
  - Ethics/HSS → PO6, PO7, PO8, PO10.
  - Projects/Internship → PO11, PO12, all PSOs.

Provide justification notes + syllabus extracts as supporting annexures